Fig. 9.6 shows the same idea in visual terms. Classically, to go from x_1 to x_2, the particle must pass through an intermediate point x. But in quantum mechanics the amplitude to go from x_1 to x_2 is an integral over all possible intermediate points.

We can carry this idea further and divide the time interval into a great many tiny intervals, as illustrated in Fig. 9.7. I won't write out the complicated formulas, but the idea should be clear. For each tiny time interval, say of size e, we include a factor

$$e^{-i\epsilon H}.$$

Then, between each pair of factors, we insert the identity so that the amplitude $C_{1,2}$ becomes a multiple integral over all the intermediate locations. The integrand is built from products of expressions with the form

$$\langle x_i | e^{-i\epsilon H} | x_{i+1} \rangle.$$

If we define $U(\epsilon)$ as

$$U(\epsilon) = e^{-i\epsilon H},$$

then we can write the entire product as

$$\langle x_2 | U^N | x_1 \rangle$$

or

$$\langle x_2 | UUUU \ldots | x_1 \rangle.$$

In this equation, U appears N times as a factor, where N is the number of epsilon steps. We can then insert identity operators between the U's.

Such an expression can be called the amplitude for the given path. But the particle does not travel along a particular path. Instead, in the limit of a large